INFOMDW: Data Wrangling

Imbalanced Data and Bias Analysis

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Topics for Today

- Imbalanced Data
- Handling Data Imbalance
- Bias Analysis

Reading Material

- Fernández, A., García, S., Galar, M., Prati, R. C., Krawczyk, B., & Herrera, F. (2018). *Learning from imbalanced data sets*. Springer. https://doi.org/10.1007/978-3-319-98074-4.

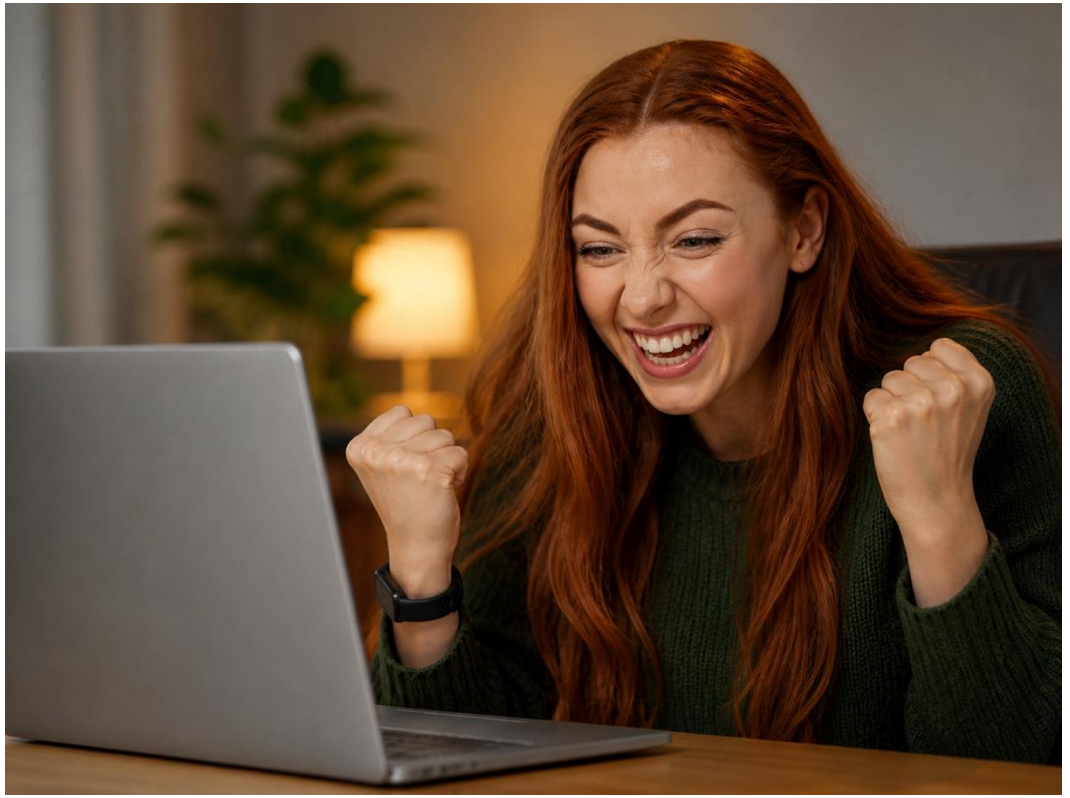
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. ACM computing surveys (CSUR), 54(6), 1-35

Imbalanced Data



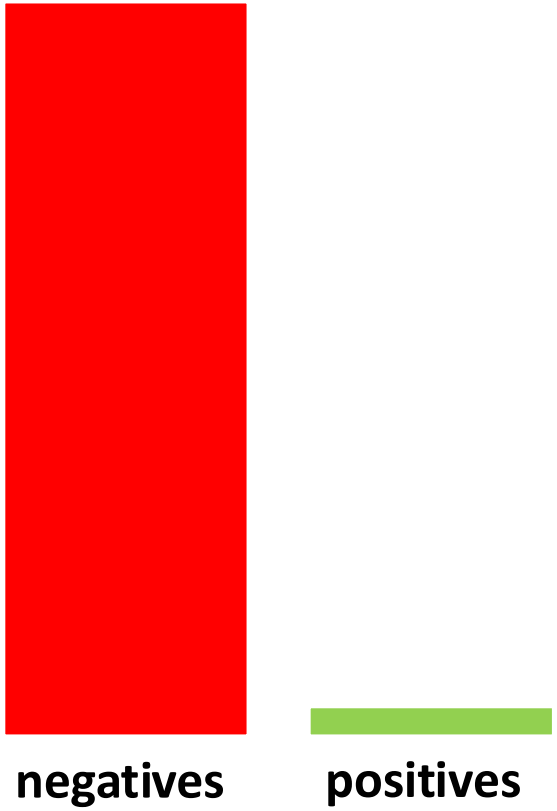
Imbalanced Data

- You trained a model to predict cancer from image data
- Your accuracy is **99.9%**



Imbalanced Data

- After plotting the class distribution
  - thousands of negative examples but just a couple of positives.





Imbalanced Data Problem

- Classifiers try to reduce the overall error (increase the accuracy) so they can be biased towards the majority class.
- Limitation of accuracy measure:
  - The problem of imbalanced classes
  - Consider a 2-class problem
    - Number of class 1 examples = 9990
    - Number of class 0 examples = 10
  - If the model predicts everything as class 1
    - $Accuracy = \frac{TP+TN}{P+N} = \frac{9990}{10000} = 99.9\%$

Your dataset is imbalanced.

Now what?

What Can We Do?

- Collect more data (difficult in many domains)
- Oversampling/undersampling of the data
- Generate synthetic data
- Adapt your learning algorithm (cost sensitive classification)

Random Over/Under Sampling

- Random oversampling: randomly duplicate data points from the minority class.
- Random under-sampling: randomly delete data points from the majority class.
- Problems:
  - Loss of information (in the case of under sampling)
  - Overfitting and fixed boundaries (over sampling)

Create Synthetic Data

SMOTE

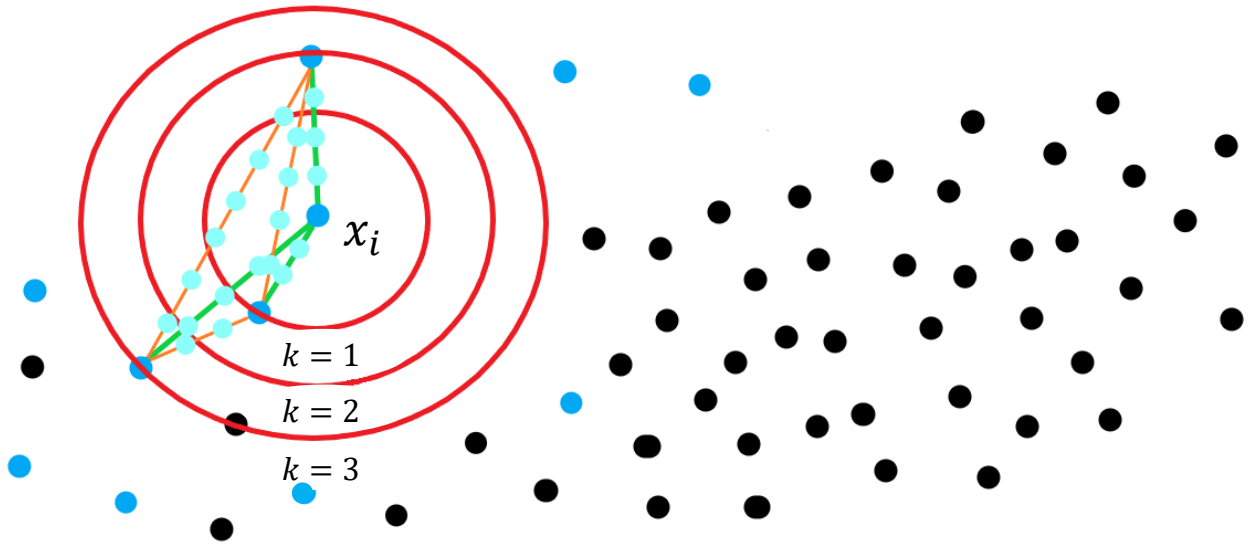
SMOTE

- Synthetic Minority Over-sampling Technique (Chawla).
- Creates new data points from the minority class.
- Operates in the feature space.

SMOTE

1. Select:
   - Choose an example from the minority class (mp1)
2. Find neighbors:
   - Identify one of its k-nearest neighbors
3. Interpolate:
   - Generate a point on the line segment that connects them

$$x_{new} = x_i + \lambda(x_j - x_i), \qquad \lambda \in [0,1]$$



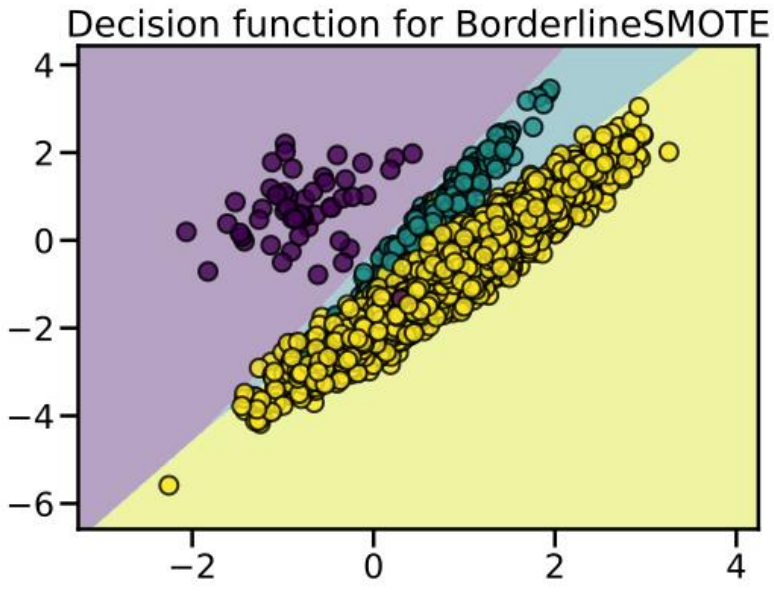
SMOTE selects a random point along the line segment between two existing points.

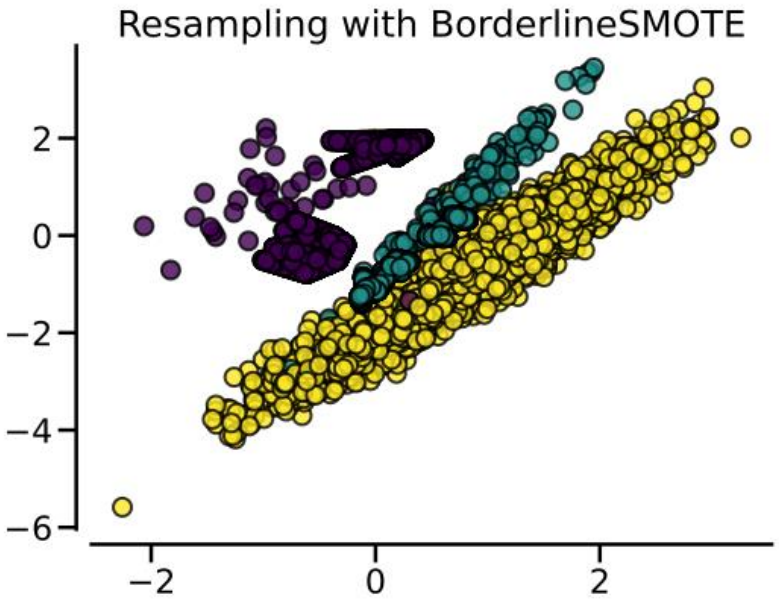
SMOTE – Things to Consider

- Do not create synthetic points on the entire dataset before splitting into train/test sets.
- Generate the synthetic examples to enrich the training data only
- **Problem with Smote:** might introduce the artificial minority class examples too deep in the majority class space.

SMOTE Variants

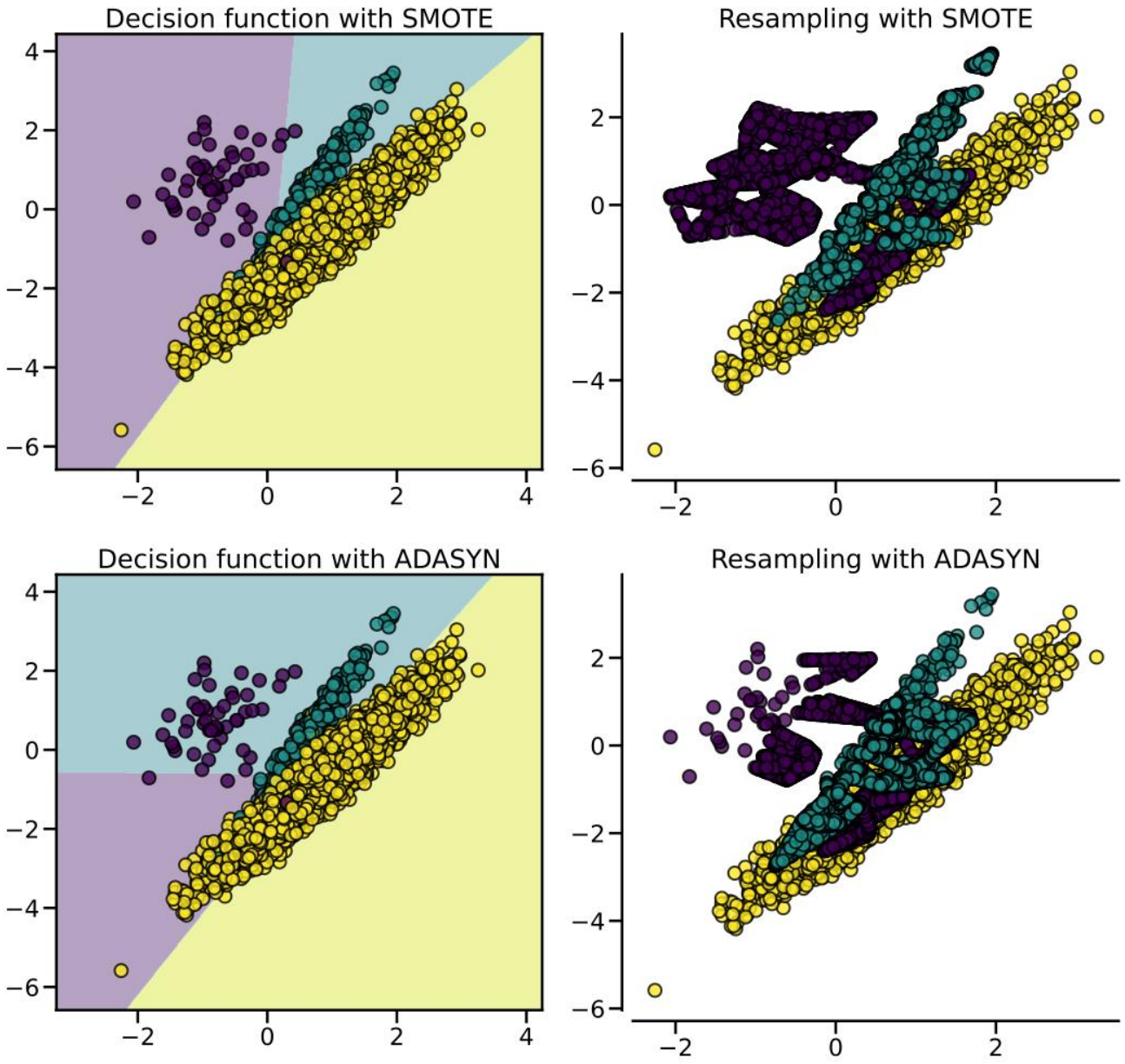
- **SMOTE-NC:** handles both numerical and categorical attributes
- **Borderline SMOTE:** focuses on minority instances near the decision boundary





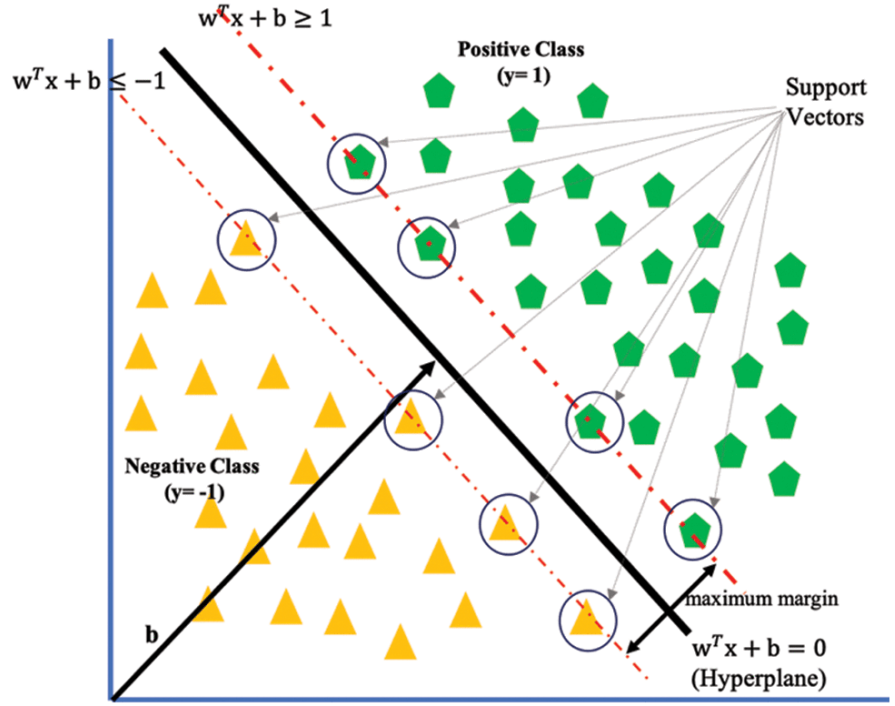
SMOTE Variants

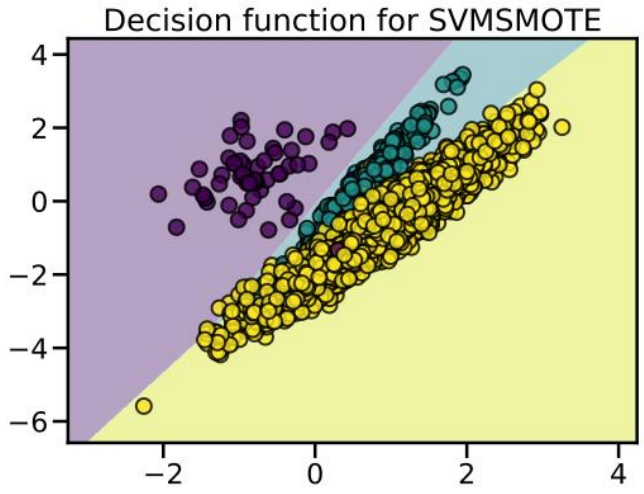
- **Adaptive Synthetic Sampling:** generates samples close to hard-to-learn minority instances

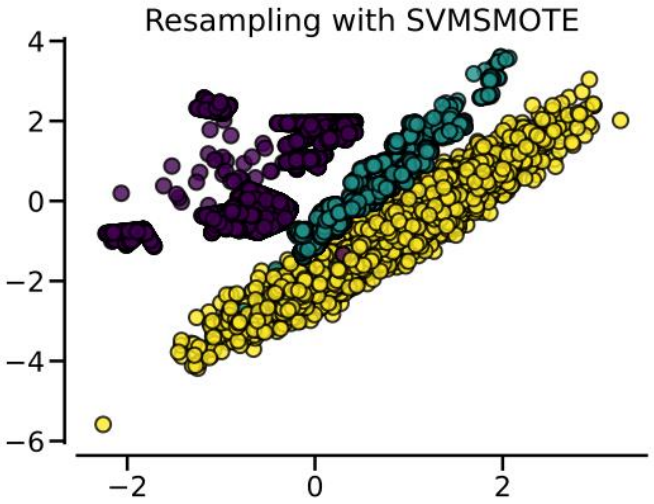


SMOTE Variants

- **SVM-SMOTE:** uses SVM to locate the vectors the supporting vectors defining the class boundary and generates examples along that margin







Yap Bee Wah, Azlan Ismail, Nur Niswah Naslina Azid, Jafreezal Jaafar, Izzatdin Abdul Aziz, Mohd Hilmi Hasan, Jasni Mohamad Zain, Machine Learning and Synthetic Minority Oversampling Techniques for Imbalanced Data: Improving Machine Failure Prediction, Computers, Materials and Continua, Volume 75, Issue 3, 2023

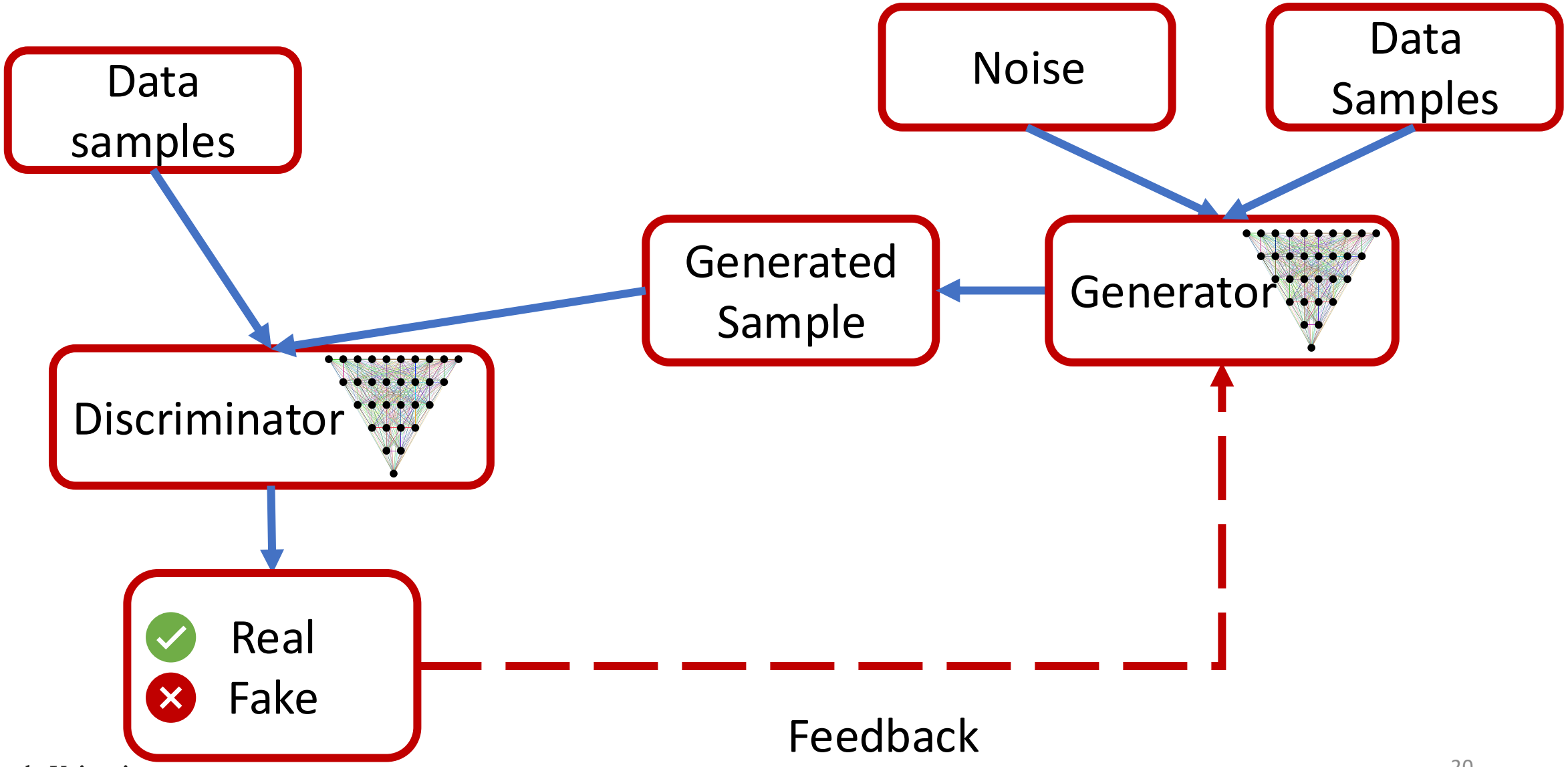
Create Synthetic Data

GANs

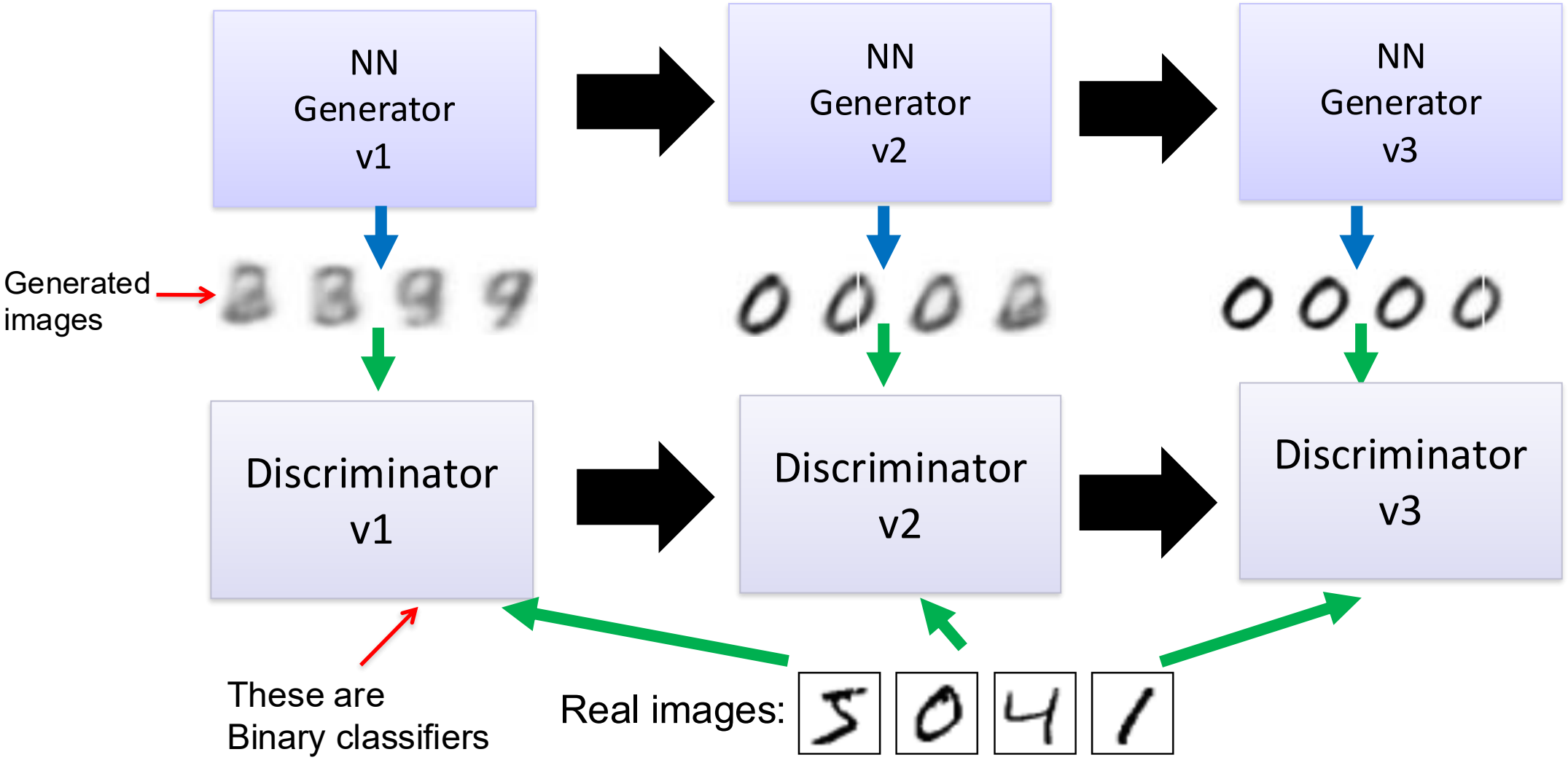
Generative Adversarial Networks (GANs)

- System of two neural networks competing against each other in a zero-sum game framework
- They were first introduced by Ian Goodfellow et al. in 2014
- Can learn to draw samples that are similar to the original examples

Generative Adversarial Networks (GANs)



Generative Adversarial Networks (GANs)



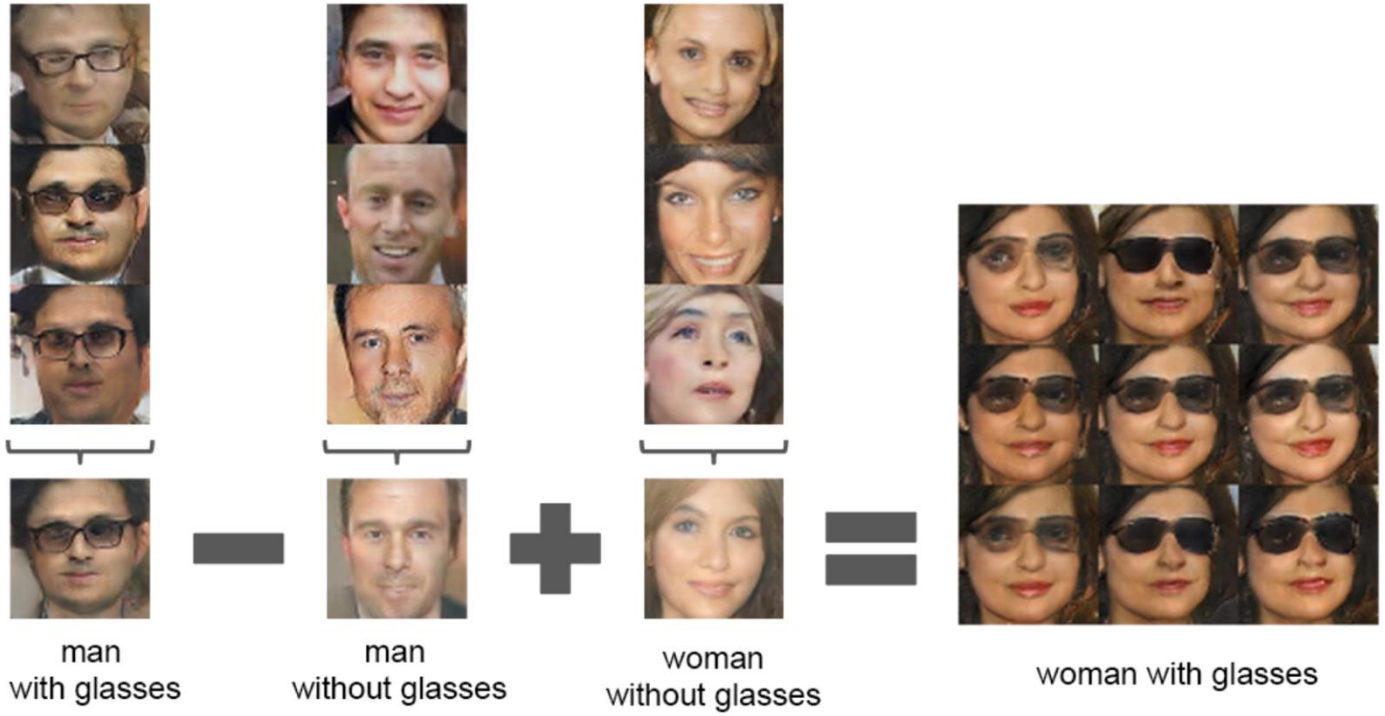
Generative Adversarial Networks (GANs)

- The **generator** tries to mimic examples from a training dataset, which is sampled from the true data distribution
  - It does so by transforming a random source of noise received as input into a synthetic sample

- The **discriminator** receives a sample, but it is not told where the sample comes from
  - Its job is to predict whether it is a real data sample or a synthetic one

How GANs Have Been Used?

- Have been used in generating images, videos, poems, some simple conversation
- Note, image processing is easy (all animals can do it), NLP is hard (only human can do it)
- This co-evolution approach might have far-reaching implications
  - This may hold the key to making computers a lot more intelligent


man
with glasses
man
without glasses
woman
without glasses
woman with glasses

How to Train GANs?

- Objective of generative network - increase the error rate of the discriminative network
  - Loss function
- Objective of discriminative network – decrease binary classification loss

Variations of the GANs

- Several new concepts built on top of GANs have been introduced –
- **InfoGAN** – Approximate the data distribution and learn interpretable, useful vector representations of data.
- **Conditional GANs (CTGAN)** - Able to generate samples taking into account external information (class label, text, another image). Force G to generate a particular type of output.

Failure Cases

- The discriminator becomes too strong too quickly and the generator ends up learning nothing
- The generator only learns very specific weaknesses of the discriminator
- The generator learns only a very small subset of the true data distribution (patterns)

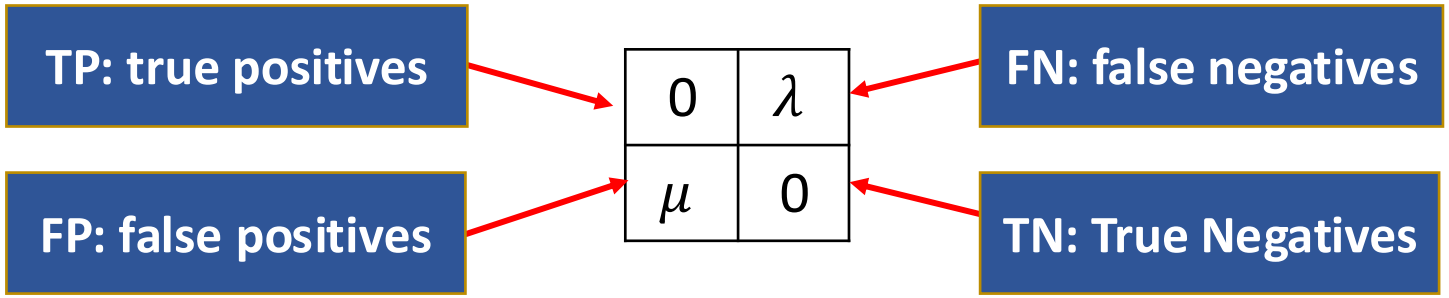
Major Problems

- Networks are difficult to converge
- Ideal goal – Generator and discriminator reach some desired equilibrium (this is rare)
- GANs are yet to converge on large problems (E.g. Imagenet).

Cost Sensitive Classification

Cost-sensitive classification

- Based on the classifier predicted probabilities.
- Binary traditional case: predict positive if probability is > 0.5
- Probability threshold can be changed using a cost matrix:



Classify as positive if: probability of positive > $\frac{\mu}{\mu+\lambda}$

Type I error
(false positive)



Type II error
(false negative)



Assign Class Weights

- In sklearn models, you can assign class weights.

fit() function in sklearn models has a class_weight parameter (see e.g., https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html).

Weights associated with classes in the form {class_label: weight}. If not given, all classes are supposed to have weight one.

e.g., class_weight = {0: 1., 1: 9.}

More Classification Evaluation Measures

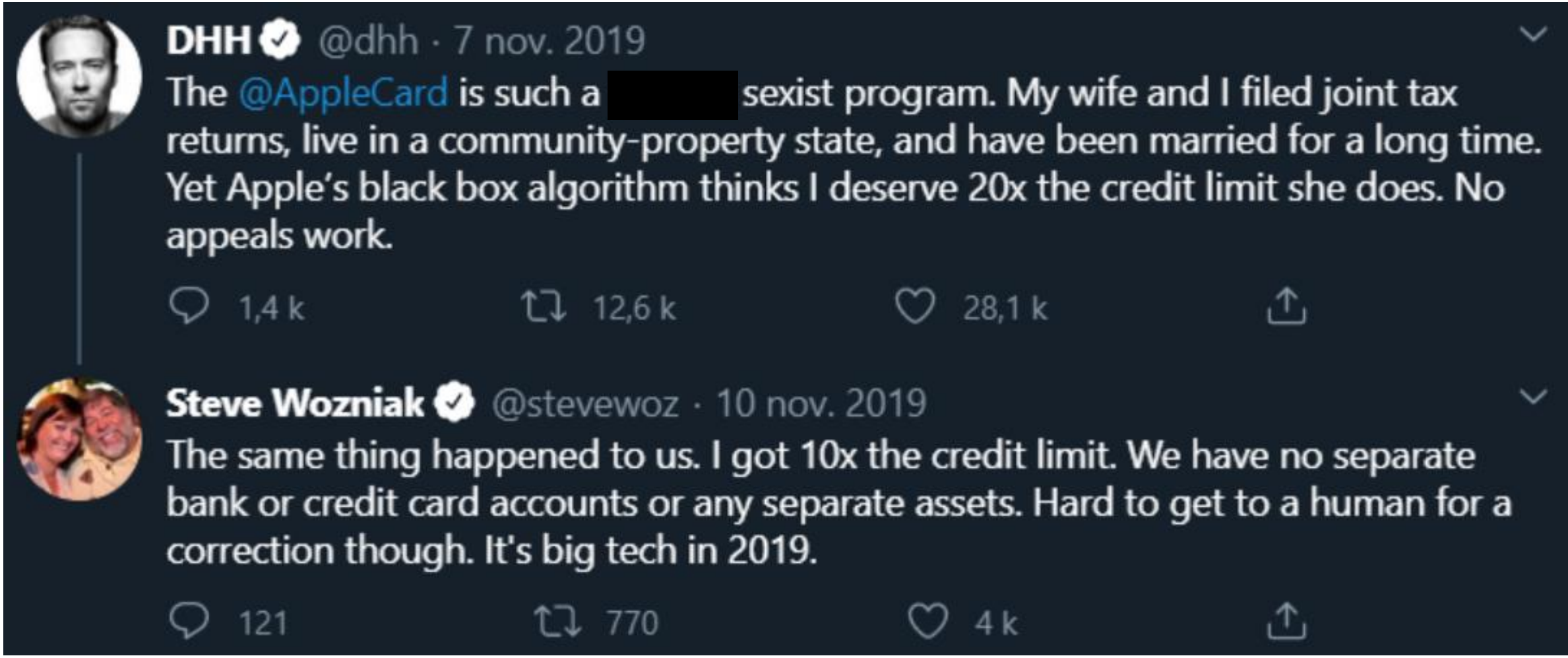
- $precision(P) = TP/(TP + FP)$
- $recall(R) = TP/(TP + FN)$
- F-measure: $\mathrm{F}_1 = (2 \times R \times P)/(R + P)$
  - In general: $\mathrm{F}_\beta = (1 + \beta^2)\,(R \times P)/(\beta^2 \times P + R)$
- $sensitivity = (TP\ /\ (TP + FN))$
- $specificity = (TN\ /\ (FP + TN))$
- $Balanced\ Accuracy = (sensitivity + specificity)\ /\ (2)$

Bias and Fairness

Unfair Applications



Unfair Algorithms – Apple Card



Unfair Algorithms – Word Embeddings in NLP



Extreme *she*	Extreme *he*
1. homemaker	1. maestro
2. nurse	2. skipper
3. receptionist	3. protege
4. librarian	4. philosopher
5. socialite	5. captain
6. hairdresser	6. architect
7. nanny	7. financier
8. bookkeeper	8. warrior
9. stylist	9. broadcaster
10. housekeeper	10. magician

Data Bias

Why ML Algorithms are Biased?

- The bias in the algorithms outcomes is not related to the way they are built

- The used dataset in training and constructing the ML models for prediction is the main reason
  - Problems comes from the bias in the training datasets

Bias in the Data

- Historical bias in the decision variable
- Limited / less informative features
- Biased data collection
- Imbalanced representation of different demographic groups



Wrap-Up

- We discussed
  - Bias and Imbalanced Data
  - Handling Class Imbalance

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